

Ensuring AI Safety, Economic Prosperity, and Global Leadership in the Age of AGI

March 15, 2025

Submission by Integral AI, Inc.

in response to RFI: Development of an Artificial Intelligence (AI) Action Plan

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.



323 Serra San Bruno
Mountain View, CA 94043
<https://www.integral.ai>



Integral AI is at the forefront of developing Artificial General Intelligence (AGI) through pioneering innovations in brain-inspired architectures, lifelong and continual learning algorithms, world models, abstraction learning, reasoning and planning, and robotics. Our innovations provide the foundation for reliable, scalable, aligned, and beneficial AGI that advances both national and global prosperity.

We believe the arrival of AGI presents an unparalleled strategic opportunity to solidify America's global leadership, advance national economic prosperity, and ensure global benefit, if—and only if—managed effectively.

To navigate the complex challenges and immense opportunities ahead, we propose a comprehensive framework for ensuring AGI is safe, reliable, aligned, and economically beneficial—without stifling innovation through excessive regulation.

Framework for AGI Safety, Alignment, and Prosperity

This framework outlines key pillars for safely developing and deploying AGI, ensuring alignment at several levels, safeguarding against misuse, and maximizing broad-based prosperity.

1. Reliability and Robustness (Behavioral Alignment)

AGI systems must exhibit robust reliability and predictable behavior, mitigating hallucinations, errors, and unintended behaviors that could compromise real-world operations.

Approach:

- Predictive World Models: Equip AGI with robust internal simulations to reason about potential consequences of planned actions, including novel or uncertain scenarios.
- Quantitative Uncertainty Estimation: Implement dynamic self-assessment so the system can flag operations with low confidence, enabling human intervention when needed.
- Continual Growth: Continuously expand and train models using newly acquired data, ensuring up-to-date knowledge and resilience against drift or adversarial inputs.

Benchmarks:

- Accuracy in predicting real-world scenarios over varying time horizons.
- Calibration of uncertainty estimation against empirical outcomes.
- Robustness metrics against adversarial attacks and unintended behavior.

2. Transparency and Auditability of AI Decisions (Developer Alignment)

Transparency is critical for trust, auditability, and accountability. Integral AI advocates for interpretability methods that do not restrict innovation yet ensure oversight capability.

Approach:

- Transparent using explainable AI techniques leveraging in mechanistic interpretability and the ability to inspect the models reasoning and planning process.
- Continuous auditable logs of model reasoning, decisions, and justifications for significant actions.

Benchmarks:

- Ability for third-party auditors to reconstruct AI reasoning steps from logged actions.
- Verifiability and explainability of decisions that have substantial societal or individual impacts.

3. Alignment with Human Intent (Individual Alignment)

To reliably execute human instructions and mitigate unintended consequences, AGI must faithfully model and adhere to human intentions.

Approach:

- Deploy AGI models capable of reasoning about intended instructions, proactively identifying unintended or unforeseen consequences.
- Establish world models capable of quantifying uncertainty to inform safe autonomous planning, particularly in critical or novel environments.

Benchmarks:

- Performance of AGI models on complex instruction-following tasks, with measurement of deviations from explicitly stated human intentions.
- Effectiveness in predicting and mitigating unintended consequences, measured through simulated scenarios involving novel tasks or unforeseen events.

4. Energy-Efficient and Scalable AGI (Resource Alignment)

Scalable AGI must be energy-efficient, drawing inspiration from biological neural computation.

Approach:

- Prioritize computational and data efficiency, enabling deployment without excessive infrastructure demands.

- Innovate in efficient learning architectures inspired by neuroscience, reducing operational costs.

Benchmarks:

- Computational performance per energy unit, benchmarked against human brain-level efficiency.
- Scalability metrics demonstrating effectiveness of AGI at large-scale deployment without linear increases in resource demands.

5. National Security and Strategic Advantage (National Alignment)

Recognizing AGI as a cornerstone of America's strategic competitiveness, we advocate securing its responsible development and deployment without imposing unnecessary restrictions or hindering innovation. Ensuring national security through AGI requires a nuanced, balanced approach rather than heavy-handed regulation or overly restrictive policies.

Approach:

- Foster robust cybersecurity practices and safeguard AGI systems against adversarial misuse, espionage, or sabotage.
- Encourage voluntary, efficient industry-government information-sharing channels designed to foster cooperation.

Benchmarks:

- Evaluation of AGI companies resilience against cyber and physical threats through regular penetration testing, red-teaming exercises, and scenario simulations involving adversarial threats.
- Capability of AGI systems to autonomously detect, respond to, and neutralize adversarial misuse attempts in real-time scenarios.

6. Common-Sense Moral Alignment (Universal Alignment)

AI should adhere to objective "common-sense" values, consistent with America's pragmatic tradition, rather than subjective moral doctrines.

Approach:

- We propose a metric termed collective agency, defined by capabilities in understanding, morally competent decision-making, ability to realize intent effectively, and sustainability or reversibility of actions.
- This ensures alignment at a universal level, preventing misuse and promoting actions aligned with common human values without imposing ideological conformity.

Benchmarks:

- Model accuracy in evaluating actions against the collective agency metric.
- Ability of AGI to mediate conflicting human requests based on objective common-sense alignment scores, demonstrating prioritization consistent with enhancing collective agency.

7. Protecting Information Integrity (Cultural Alignment)

AGI must not exacerbate cultural fragmentation or disrupt societal cohesion through misinformation or psychological manipulation.

Approach:

- Develop algorithms inherently resistant to manipulation, adversarial misinformation campaigns, and cognitive exploitation.
- Ensure AI recommendation engines avoid polarization or biased filtering by prioritizing balanced and verified information.

Benchmarks:

- Resistance of AI systems to adversarial content generation and recommendation manipulation.
- Ability of AGI to autonomously detect and mitigate misinformation spread across digital platforms and social networks.

8. Shared Prosperity and Economic Inclusivity (Economic Alignment)

To avoid the pitfalls of earlier industrial disruptions, we propose fundamentally reimagining economic incentives and distribution mechanisms.

Our economic framework, described comprehensively in “The Rise of Superintelligence”, emphasizes equitable distribution of AGI-generated prosperity, reflecting true value creation including externalities.

Approach:

- Develop novel economic measures reflecting positive and negative externalities (environmental, societal, cultural) created by AI-generated goods and services.
- Facilitate broad societal distribution of AGI-derived wealth, ensuring human flourishing.

Benchmarks:

- Economic inclusivity metrics demonstrating tangible improvement in shared prosperity outcomes across diverse demographics.
- externality pricing effectiveness benchmarks, evaluating alignment between AGI-generated prosperity and societal flourishing metrics.

Preliminary Policy Recommendations

The rapid advancement of AGI presents both opportunities and challenges that demand proactive, adaptive, and continuously evolving policy measures. The following preliminary recommendations highlight key areas where government-private sector collaboration is essential to ensuring AGI development remains secure, socially aligned, and beneficial.

1. Public-Private Partnerships for AI Security

- Bolster robust security “best practices” for AGI labs, including physical security, access controls for data centers, and threat-intelligence sharing channels.
- Incentivize businesses to adopt new security standards through streamlined compliance processes and procurement advantages.

2. AGI Security Standards and Audits

- Develop industry-led security and reliability standards for large-scale or frontier AGI models; require periodic “stress tests” to evaluate worst-case adversarial uses.
- Encourage red-teaming as a formal discipline within AI organizations, in partnership with government agencies and third-party experts.

3. AI Safety Validation Markets

- Establish competitive “safety testing marketplaces” that reward independent researchers and vendors for identifying system vulnerabilities.
- Enable flexible cost-sharing models in which private actors can invest in external audits to meet public or federal compliance needs.

4. Tools for AI Forensics and Explainability

- Fund the development and standardization of forensic toolkits that reconstruct critical decision pathways in AGI models.
- Require that high-impact systems integrate logging features that support post-hoc incident investigation by authorized government agencies or third-party auditors.

5. AGI Readiness Stress Tests

- Mandate recurring large-scale security and resilience drills for core AI infrastructure—particularly in contexts with potential systemic impacts (financial markets, critical infrastructure, defense).
- Collaborate with data-center operators, hardware vendors, and cloud providers to ensure resilience amid unexpected compute or energy disruptions.

6. National AI Compute Forecasting

- Task a cross-agency group with projecting short- and long-term demand for AI-related energy and compute resources, enabling forward-looking investments and policies.

- Encourage state-level coordination to site data centers in regions with abundant clean energy, simplifying environmental reviews.

7. Strengthen Domestic AI Supply Chain

- Expand incentives for onshore semiconductor manufacturing and next-generation computing R&D.
- Partner with private industry to identify potential single points of failure (e.g., raw material dependencies) that could disrupt AGI deployment timelines.

We are actively working on more comprehensive recommendations and believe that closer collaboration with government stakeholders will be essential in shaping a national strategy that fosters innovation, strengthens security, and maximizes societal benefit.

Conclusion

Integral AI's framework positions the United States to responsibly lead AGI development, harnessing its benefits while addressing its risks comprehensively yet flexibly. Rather than burdensome overregulation, our approach champions innovation-friendly but accountable AGI deployment.

We believe AGI can enhance American economic prosperity, preserve strategic advantage, and contribute significantly to global human flourishing when guided by common-sense alignment principles.

We look forward to working closely with the administration and policymakers to advance America's leadership in AGI while ensuring a prosperous and stable future for all.